

SE Lab 1: Requirements Engineering & UML

Use-Case Modelling

PROBLEM STATEMENT #39 | Retail, E-Commerce & Finance

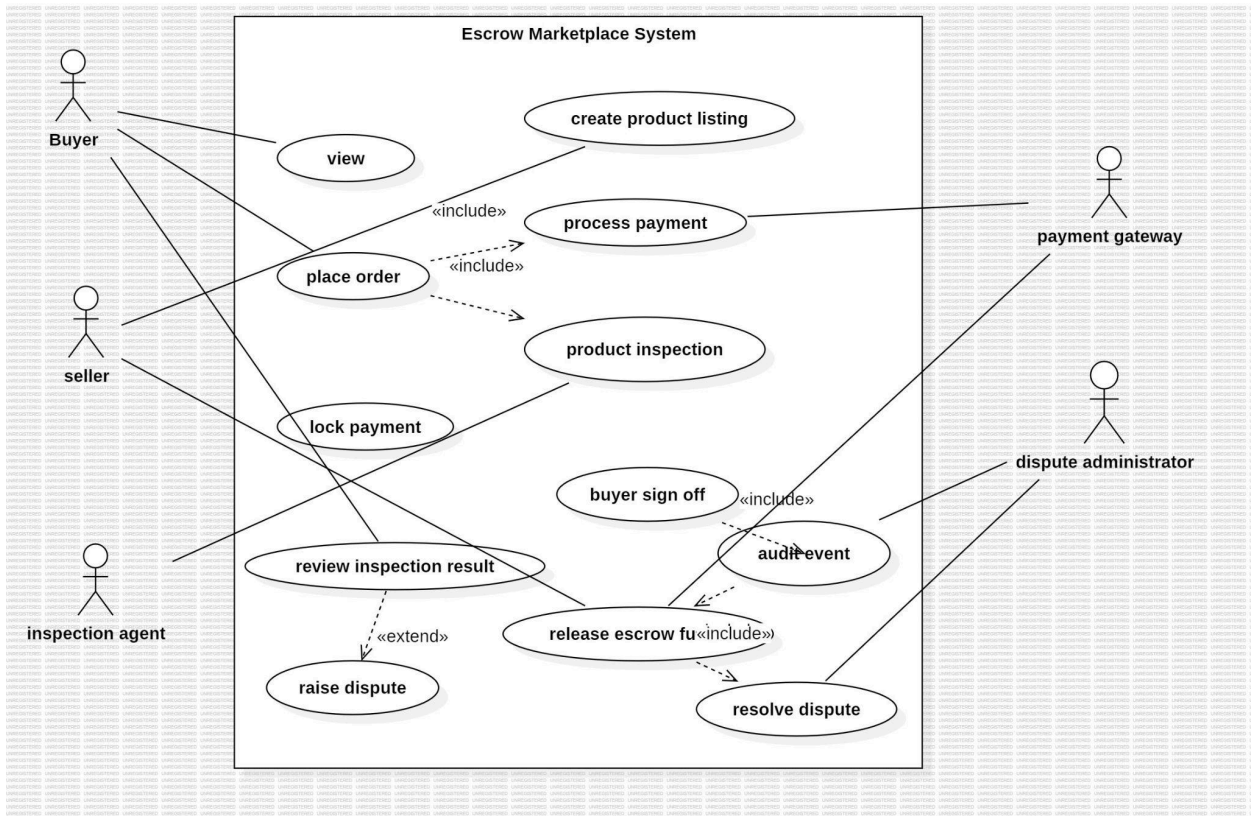
Second-Hand Product Escrow Marketplace

Deliverable 1: Complete Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall lock buyer payment in escrow and transition the order state to 'Under Inspection' until an inspector submits physical verification sign-off.	High	Pass: After successful payment, the order state becomes 'Under Inspection' and escrow funds remain locked until inspection sign-off and buyer sign-off. Fail: funds are released before both approvals.	Protects both parties by preventing premature payment release during inspection.
FR-002	Functional	The system shall allow a seller to create a product listing containing product details, asking price, condition information, and inspection requirements.	High	Pass: A listing cannot be published unless all mandatory fields are completed; a published listing displays the entered details and price. Fail: incomplete mandatory data is published.	Provides the buyer with sufficient information to evaluate the product before purchase.
FR-003	Functional	The system shall allow a buyer to select an available product, confirm the order, and authorize payment into the escrow account.	High	Pass: A valid order creates an order record and payment transaction, and the escrow status becomes 'Locked'. Fail: an order is marked confirmed without a successful payment authorization.	Starts the controlled transaction and establishes the escrow protection.
FR-004	Functional	The system shall allow an inspection agent to record inspection findings, upload verification evidence, and submit a physical verification sign-off for an order.	High	Pass: The inspection record stores findings, evidence, agent identity, timestamp, and sign-off status; the order cannot proceed to buyer sign-off without a submitted inspection. Fail: an incomplete inspection is accepted.	Creates an auditable independent condition assessment.
FR-005	Functional	The system shall allow the buyer to review the inspection result and either provide sign-off to release escrow funds or raise a dispute.	High	Pass: Buyer approval moves the order to 'Approved for Release' and triggers escrow release; rejection creates an active dispute and keeps escrow locked. Fail: funds are released after a buyer dispute.	Ensures final release depends on buyer acceptance and provides protection when the product does not match expectations.
NFR-001	Nonfunctional	The system shall freeze escrow funds immediately when a dispute-resolution ticket is created and	High	Pass: In tests, a dispute changes escrow status to 'Frozen' before any release/withdrawal operation can complete.	Prevents financial loss while a transaction is being investigated.

		shall prevent withdrawals from the disputed transaction during active investigation.		Fail: any disputed transaction releases or withdraws funds during investigation.	
NFR-002	Nonfunctional	The system shall record all escrow state changes, inspection submissions, buyer sign-offs, dispute actions, and fund-release events in an auditable log with a timestamp and actor identifier.	Medium	Pass: Every tested transaction event produces one immutable audit entry containing event type, actor ID, timestamp, and order ID. Fail: any tested state-changing event has no corresponding audit entry.	Supports traceability, accountability, and dispute investigation.

Deliverable 2: UML Use-Case Diagram:



Deliverable 3: Use-Case Flow Specification:

Use Case ID	UC-03
Use Case	Complete Escrow Purchase
Primary Actor	Buyer
Supporting Actors	Inspection Agent, Payment Gateway, Seller
Goal	Purchase a second-hand product safely, obtain independent inspection, and release escrow only after buyer approval.

Preconditions

- The buyer is authenticated and the selected product listing is available.
- The seller has published the listing with the required product and condition information.
- The payment gateway is available to authorize the buyer's payment.

Postconditions

Successful path: inspection is recorded, buyer signs off, escrow is released to the seller, and the order is marked completed.

- Alternate path: if the buyer disputes the result, escrow remains frozen and a dispute ticket is created for investigation.

Main Success Scenario

1. Buyer selects an available product and chooses "Purchase".
2. System displays the order summary, seller details, price, and escrow terms.
3. Buyer confirms the order and selects a payment method.
4. System sends the payment authorization request to the Payment Gateway.
5. Payment Gateway authorizes the payment.
6. System locks the authorized amount in escrow and changes the order state to "Under Inspection".
7. System assigns/notifies an Inspection Agent for the physical verification.

8. Inspection Agent examines the product, records findings, uploads evidence, and submits physical verification sign-off.
9. System stores the inspection record and notifies the Buyer that the result is ready.
10. Buyer reviews the inspection result and evidence.
11. Buyer provides sign-off accepting the product condition.
12. System changes the order state to “Approved for Release”, releases the escrow funds to the Seller, records the release event in the audit log, and marks the order completed.

Alternate Flow — Buyer Raises a Dispute

- AF-1. At Step 11, the Buyer rejects the inspection result and selects “Raise Dispute”.
- AF-2. System immediately freezes the escrow amount and prevents release/withdrawal for the disputed transaction.
- AF-3. System creates a dispute-resolution ticket containing the order ID, inspection result, buyer reason, and relevant evidence.
- AF-4. A dispute administrator reviews the case and resolves it according to marketplace policy.
- AF-5. If the dispute is upheld, the system processes the approved resolution without releasing funds prematurely; if rejected, the system may return the transaction to the buyer-approval path.
- AF-6. The system records each dispute and escrow state transition in the audit log.